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SEAT No. _____

No. of Printed Pages: 2

SARDAR PATEL UNIVERSITY

M. Sc. I. T. (Integrated) Examination, 7th Semester

Tuesday, 30th October, 2018

PS07EIT01: Image Processing

Time: 02:00 PM to 04:00 PM

Total Marks: 70

Q:1 Give answers of following Multiple Choice Questions

[08]

[01] What is image?

- (A) Picture
- (B) Matrix of pixel
- (C) Collection of pixel
- (D) All of these

[02] The process of extracting information from the image is called as

- (A) Image enhancement
- (B) Image restoration
- (C) Image Analysis
- (D) Image compression

[03] Continuous functions are sampled to form a

- (A) Fourier series
- (B) Fourier transform
- (C) Fast Fourier series
- (D) Digital image

[04] Image can be blurred using

- (A) Low pass filter
- (B) Contouring
- (C) Erosion
- (D) High pass filter

[05] Square of standard deviation is called

- (A) Variance
- (B) Noise
- (C) Restoration
- (D) Power

[06] The purpose of restoration is to gain

- (A) Degraded Image
- (B) Original Image
- (C) Pixels
- (D) Coordinates

[07] Coding redundancy works on

- (A) Pixels
- (B) Matrix
- (C) Intensity
- (D) Coordinates

[08] In the image $M \times N$, M is

- (A) Rows
- (B) Column
- (C) Level
- (D) Intensity

Q:2 Answer the following short questions (any Seven)

[14]

[01] Define: Digital Image, Pixel.

[02] List out first five fundamental steps of DIP.

[03] Write applications of image transform.

[04] Define: Time domain, Frequency domain

[05] What is image enhancement?

[06] Define noise and list out types for source of noise.

[07] What is periodic noise?

[08] What is Psychovisual redundancy?

[09] List-out fundamental coding theorems.

(P. T. O.)

- Q:3** [A] Explain any three applications of Image Processing [06]
[B] Explain fundamental steps in image processing [06]
- OR
- [B] Explain Elements of visual perception. [06]
- Q:4** [A] Explain Fourier Transform and Inverse Fourier Transform. [06]
[B] What is contrast intensification? Explain Histogram specification & Local contrast stretching. [06]
- OR
- [B] Explain Edge-Preserving smoothing and Low-Pass filtering. [06]
- Q:5** [A] Explain Linear, Position-Invariant Degradations. [06]
[B] Explain Estimating the Degradation Function. [06]
- OR
- [B] Explain Constrained Least Squares & Geometric Mean filter. [06]
- Q:6** [A] Explain channel encoder & decoder. [06]
[B] Explain Transform coding. [06]
- OR
- [B] Explain variable length coding in brief. [06]

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